lecture 5

24p-0535 OLT

Collection of Quizes.

Predicates

$P(x) = x + 5 = 8$ → predicate $x \in \{1,2,3,4\}$.

Subject.

$1 + 5 = 8$ F
$2 + 5 = 8$ F
$3 + 5 = 8$ T
$4 + 5 = 8$. F.

1) Subject.
2) predicate (Condition).

$y = f(x)$.

Range. Domain.

Ex1 P31

$P(x) = x > 3$ $p(4)$ & $p(2)$.

$P(2) = 2 > 3$ F
$P(4) = 4 > 3$ T.

Ex2 P31

$A(x) =$ Computer x is Under attack.

Given:- CS1 & MATH1 are Under attack.

$A(CS1) =$ Computer CS1 is under attack $= T$
$A(CS2) =$ " CS2 " " " $= F$.

Ex3:- P31

$Q(x, y) = x = y + 3$
$Q(1, 2) = 1 = 2 + 3 = F$.
$Q(3, 0) = 3 = 0 + 3 = T$.

Quantifiers.

Universal $\forall$

$\forall x P(x) = P(1) \wedge P(2) \wedge \dots \wedge P(N)$. $x \in \{1,2,3,\dots N\}$.

Counter Example.

"for all", "for every", "All of" ---- p 33.

Ex 8 $P(x) = x+1 > x$ $x \in \mathbb{R}$.

P 33 $\forall x P(x) = T$

Ex 10 :- $P(x)$ $x^2 > 0$ $x \in \mathbb{Z}$.

P 34 $\forall x P(x) = F$

Counter Example $P(0) = 0^2 > 0 = F$.

Ex 11 :- $P(x) = x^2 < 10$ $x \in \{1,2,3,4\}$.

P 34 $\forall x P(x) = P(1) \wedge P(2) \wedge P(3) \wedge P(4)$.

$= 1^2 < 10 \wedge 2^2 < 10 \wedge 3^2 < 10 \wedge 4^2 < 10$.

$= T \wedge T \wedge T \wedge F = F$.

Existential $\exists$

$\exists x P(x) = P(1) \vee P(2) \vee \dots \vee P(N)$. $x \in \{1,2,3,\dots,N\}$.

for atleast one, there is, for some, --- p 35.

Ex 14 $\exists x P(x) = x > 3$ $x \in \mathbb{R}$.

P 35 $P(4) = 4 > 3 = T$.

Ex 15 $P(x) = x = x+1$ $x \in \mathbb{R}$.

P 35 $\exists x P(x) = F$.

Ex 17 :- $\forall x<0 \ (x^2>0)$ $x \in \mathbb{R}$.
P 36 let $p(x) = x^2>0$.
$\forall x<0 \ p(x) = ?$

Negating Quantifiers.

$\neg \forall x\, p(x). = \neg\left(p(1)\wedge p(2)\wedge\cdots\wedge p(N)\right).$ $x \in \{1,2,3,\cdots N\}.$
$= \neg p(1) \vee \neg p(2) \vee \neg p(3) \vee \cdots \vee \neg p(N).$

$= \exists x\, \neg p(x).$ — (1)

De Morgan's
$\neg(p\wedge q) = \neg p \vee \neg q.$

$(p_1 \wedge p_2 \wedge p_3 \wedge \cdots \wedge p_N).$
$\neg p_1 \vee \neg p_2 \vee \cdots \vee \neg p_N.$

$\neg \exists x\, p(x) = \neg\left(p(1) \vee p(2) \vee \cdots \vee p(N)\right).$ $x \in \{1,2,3,\cdots N\},$

$= \neg p(1) \wedge \neg p(2) \wedge \neg p(3) \wedge \cdots \wedge \neg p(N).$
$= \forall x\, \neg p(x).$ — (2)

$\neg \forall x\, p(x) = \exists x\, \neg p(x)$ — (1).
$\neg \exists x\, p(x) = \forall x\, \neg p(x)$ — (2).

Find Negation.

$\neg\left(\neg \forall z\, \exists y\, \forall x\, \neg p(x,y,z)\right)$

$\forall z\, \exists y\, \forall x\, \neg p(x,y,z)$

$\neg \forall x\, p(x) = \exists x\, \neg p(x)$ — (1).
$\neg \exists x\, p(x) = \forall x\, \neg p(x)$ — (2).

$p(z).$
$\forall z\ \boxed{\exists y\, \forall x\, \neg p(x,y,z)}$ Find Negation.

$$\neg \forall_z \overbrace{\boxed{\exists y \, \forall x \neg P(x,y,z)}}^{P(z).}$$

Find Negation.

$$= \exists z \neg \exists y \overbrace{\boxed{\forall x \neg P(x,y,z)}}^{P(y).}$$

$$= \exists z \forall y \neg \forall x \neg P(x,y,z).$$

$$= \exists z \forall y \exists x \neg \neg P(x,y,z).$$

$$= \exists z \forall y \exists x \, P(x,y,z).$$

$\neg \forall_x P(x) =$ $\quad x+1=0$ $\quad x \in \{0,-1,+1\}.$

$\exists_x \neg P(x) = \neg P(0) \vee \neg P(+1) \vee \neg P(-1).$

$= 0+1 \neq 0 \vee 1+1 \neq 0 \vee -1+1 \neq 0.$

$\quad T \vee T \vee F = T.$

Quiz 1:- $\qquad$ 11-Sep-2025.

$\exists_x P(x) = x \in B.$ $\qquad x \in \{a,b,c\}.$

$B = \{a,b,c,d\}.$

Solution:-

$\exists_x P(x) = P(a) \vee P(b) \vee P(c).$

$= a \in B \vee b \in B \vee c \in B.$

$= T \vee T \vee T = T.$